Lab Assignment 10

Title: Data Visualization III

PROBLEM STATEMENT:

Download the Iris flower dataset or any other dataset into a DataFrame.(e.g., https://archive.ics.uci.edu/ml/datasets/Iris). Scan the dataset and give the inference as:

1. List down the features and their types (e.g., numeric, nominal) available in the dataset.

2. Create a histogram for each feature in the dataset to illustrate the feature distributions.

3. Create a boxplot for each feature in the dataset.

4. Compare distributions and identify outliers.

THEORY:

Histogram:-

Pandas.DataFrame.hist() function is useful in understanding the distribution of numeric variables. This function splits up the values into the numeric variables. Its main functionality is to make the Histogram of a given Data frame.

The distribution of data is represented by Histogram. When Function Pandas DataFrame.hist() is used, it automatically calls the function matplotlib.pyplot.hist() on each series in the DataFrame

Syntax: DataFrame.hist(data, column=None, by=None, grid=True, xlabelsize=None, xrot=None, ylabelsize=None, yrot=None, ax=None, sharex=False, sharey=False, figsize=None, layout=None, bins=10, backend=None, legend=False, **kwargs)
Parameters:
data: DataFrame
column: str or sequence
xlabelsize: int, default None
ylabelsize: int, default None
ax: Matplotlib axes object, default None
**kwargs
All other plotting keyword arguments to be passed to matplotlib.pyplot.hist().
Return:
matplotlib.AxesSubplot or numpy.ndarray

Box Plot :-

Box Plot is the visual representation of the depicting groups of numerical data through their quartiles. Boxplot is also used for detect the outlier in data set. It captures the summary of the data efficiently with a simple box and whiskers and allows us to compare easily across groups. Boxplot summarizes a sample data using 25th, 50th and 75th percentiles. These percentiles are also known as the lower quartile, median and upper quartile.

A box plot consist of 5 things.

- Minimum
- First Quartile or 25%
- Median (Second Quartile) or 50%
- Third Quartile or 75%
- Maximum

Draw the boxplot using seaborn library:

Syntax : seaborn.boxplot(x=None, y=None, hue=None, data=None, order=None, hue_order=None, orient=None, color=None, palette=None, saturation=0.75, width=0.8, dodge=True, fliersize=5, linewidth=None, whis=1.5, notch=False, ax=None, **kwargs)

Parameters:
x = feature of dataset
y = feature of dataset
hue = feature of dataset
data = dataframe or full dataset
color = color name

Identify outliers:-

Detect and Remove the Outliers using Python

An Outlier is a data-item/object that deviates significantly from the rest of the (so-called normal)objects. They can be caused by measurement or execution errors. The analysis for outlier detection is referred to as outlier mining. There are many ways to detect the outliers, and the removal process is the data frame same as removing a data item from the panda's data frame.

Here pandas data frame is used for a more realistic approach as in real-world project need to detect the outliers arouse during the data analysis step, the same approach can be used on lists and series-type objects.

Detecting the outliers

Outliers can be detected using visualization, implementing mathematical formulas on the dataset, or using the statistical approach. All of these are discussed below.

1. Visualization

- **Using Box Plot**
- **Using ScatterPlot**
- **Z-score**

Z- Score is also called a standard score. This value/score helps to understand that how far is the data point from the mean. And after setting up a threshold value one can utilize z score values of data points to define the outliers.

Zscore = (data_point -mean) / std. deviation

- **IQR (Inter Quartile Range)**

Inter Quartile Range approach to finding the outliers is the most commonly used and most trusted approach used in the research field.

IQR = Quartile3 – Quartile1

Procedure & Code:-

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
data1 = pd.read_csv("C:/Users/Admin/Downloads/Iris.csv")
data1.head()
```

Out[2]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

In [3]:
```
print(data1.columns)
```

```
Index(['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',
       'Species'],
      dtype='object')
```

In [7]:
```
data1.info()
```

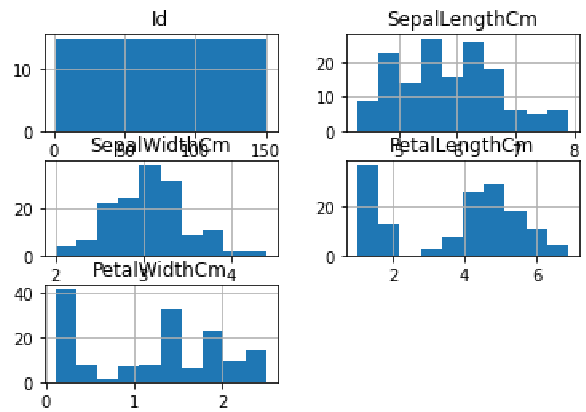
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   Id             150 non-null    int64  
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object 
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

In [13]:
```
data1.dtypes
```

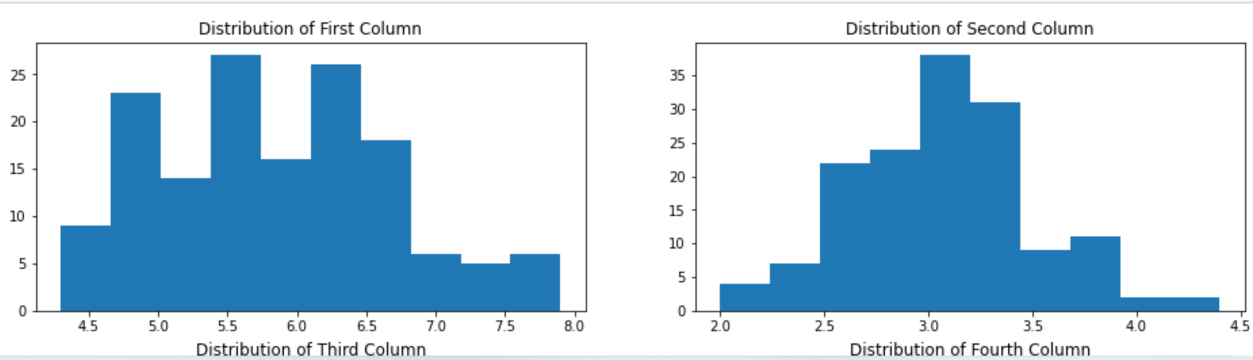
Out[13]:
```
Id                 int64
SepalLengthCm    float64
SepalWidthCm     float64
PetalLengthCm    float64
PetalWidthCm     float64
Species           object
dtype: object
```

In [20]:
```python
data1.hist()
```

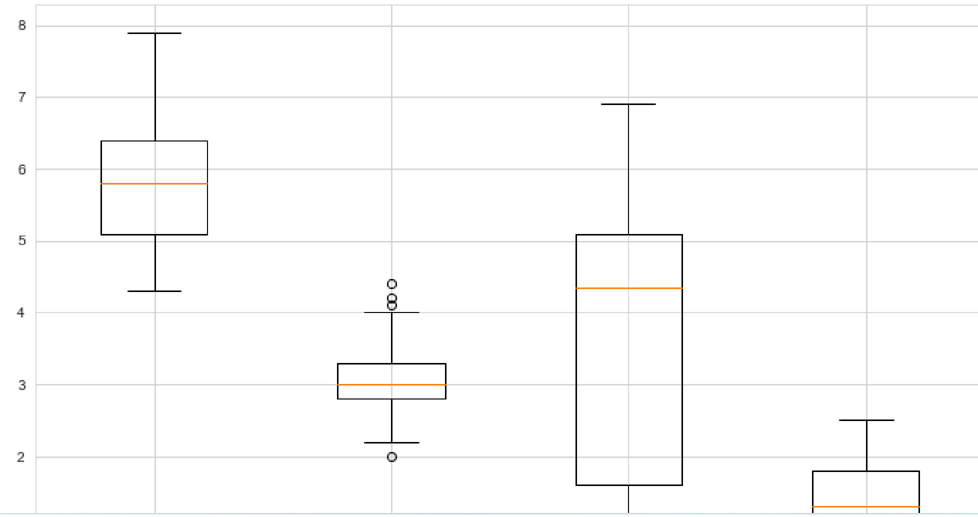
Out[20]:
```
array([[<AxesSubplot:title={'center':'Id'}>,
        <AxesSubplot:title={'center':'SepalLengthCm'}>],
       [<AxesSubplot:title={'center':'SepalWidthCm'}>,
        <AxesSubplot:title={'center':'PetalLengthCm'}>],
       [<AxesSubplot:title={'center':'PetalWidthCm'}>, <AxesSubplot:>]],
      dtype=object)
```



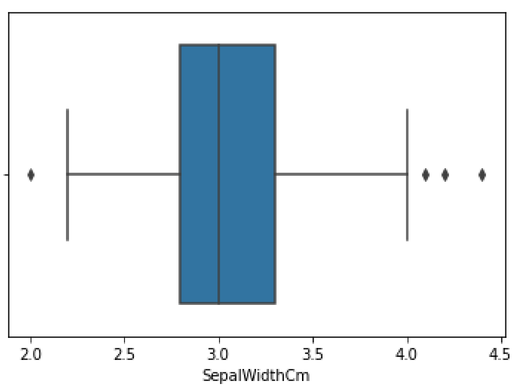
In [17]:
```python
fig, axes = plt.subplots(2, 2, figsize=(16, 8))

axes[0,0].set_title("Distribution of First Column")
axes[0,0].hist(data1["SepalLengthCm"]);

axes[0,1].set_title("Distribution of Second Column")
axes[0,1].hist(data1["SepalWidthCm"]);

axes[1,0].set_title("Distribution of Third Column")
axes[1,0].hist(data1["PetalLengthCm"]);

axes[1,1].set_title("Distribution of Fourth Column")
axes[1,1].hist(data1["PetalWidthCm"]);
```



```
In [39]: data_to_plot = [data1["SepalLengthCm"],data1["SepalWidthCm"],data1["PetalLengthCm"],data1["PetalWidthCm"]]
         # Creating a figure instance
         fig = plt.figure(1, figsize=(12,8))
         # Creating an axes instance
         ax = fig.add_subplot(111)
         # Creating the boxplot
         bp = ax.boxplot(data_to_plot);
```




```
In [19]: sns.boxplot(data1['SepalWidthCm'])

C:\ProgramData\Anaconda3\lib\site-packages\seaborn\_decorators.py:36: FutureWarning: Pass the following variable
rg: x. From version 0.12, the only valid positional argument will be `data`, and passing other arguments without
yword will result in an error or misinterpretation.
  warnings.warn(

Out[19]: <AxesSubplot:xlabel='SepalWidthCm'>
```




```
In [36]: print(np.where(data1['SepalWidthCm']>4.0))

(array([15, 32, 33], dtype=int64),)
```

CONCLUSION:

We have successfully implemented operations on the ‘iris’ dataset, also we have successfully plotted a histogram, boxplot and identified outliers.